Lesson 3.7 — The Remainder Theorem

Big idea: If you divide a polynomial P(x) by (x – c), the remainder is exactly P(c). That's a one-line shortcut for evaluating polynomials.

Quick review

Statement: $P(x) = (x - c)Q(x) + R$, where $R = P(c)$.
Use 1: evaluate P(c) by doing synthetic division with c — the last entry is the answer.
Use 2: check whether (x – c) is a factor of P. If P(c) = 0, yes.
Caveat: the divisor must be a linear factor of the form (x – c). For (x + 3), use c = –3.

Worked example

Worked example. Use the Remainder Theorem to find P(2) where $P(x) = 3x^3 - 5x^2 + 4x - 1$.

Synthetic with c = 2: row from coefficients 3, –5, 4, –1 → 6, 2, 12; sums 3, 1, 6, 11. Last entry 11 = P(2).

Warm-ups

1. If P(x) divided by (x – 4) gives remainder 9, what is P(4)?
2. Use the Remainder Theorem: find P(–1) for $P(x) = x^3 + 2x^2 - 5x + 1$.
3. Is (x – 3) a factor of $P(x) = x^2 - 9$?

Practice

1. Find P(–2) for $P(x) = 2x^4 - 3x^2 + x + 5$ using synthetic division.
2. Determine whether (x + 1) is a factor of $P(x) = x^3 - 4x^2 + x + 6$.
3. If $P(x) = x^3 + ax + 2$ and (x – 1) is a factor, find a.

Challenge

1. Find a so that $P(x) = 2x^3 - 5x^2 + ax + 6$ leaves remainder 12 when divided by (x – 3).
2. Use the Remainder Theorem to find a polynomial P(x) of degree 3 satisfying P(0) = 6, P(1) = 0, P(2) = 0, P(–1) = 0.

Answer key — Lesson 3.7

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $P(4) = 9$.
2. $(-1)^3 + 2(1) - 5(-1) + 1 = -1 + 2 + 5 + 1 = 7$.
3. $P(3) = 0 \Rightarrow$ yes, $(x - 3)$ is a factor.

Practice

1. Synthetic with –2 on (2, 0, –3, 1, 5): –4, 8, –10, 18; sums 2, –4, 5, –9, 23. $P(-2) = 23$.
2. $P(-1) = -1 - 4 - 1 + 6 = 0 \Rightarrow$ yes, $(x + 1)$ is a factor.
3. $P(1) = 1 + a + 2 = 0 \Rightarrow a = -3$.

Challenge

1. $P(3) = 54 - 45 + 3a + 6 = 15 + 3a$. Set $= 12 \Rightarrow a = -1$.
2. Zeros at 1, 2, –1: $P(x) = a(x - 1)(x - 2)(x + 1)$. $P(0) = a(-1)(-2)(1) = 2a = 6 \Rightarrow a = 3$. $P(x) = 3(x - 1)(x - 2)(x + 1)$.